

Sri Lanka Institute of Information Technology



2025-Y2-S1-MLB-WE1G2-08

Web Based Transportation System

SE2030 - Software Engineering

B.Sc. (Hons) in Information Technology

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Scenario: Transport Management System

User Stories for each persona with Transport Management System

Identify Personas:

- User - Book rides, views ride history, updates profile and choose payment options.
- Driver - Accept ride requests, view ride details and updates ride status.
- Admin - Oversees users, manages driver registrations and checks reports.

Persona: User

User Stories:

1. **As a User**
I want to book a ride
So that I can travel to my destination.
2. **As a User**
I want to see my rides
So that I can view my current rides and previous rides.
3. **As a User**
I want to go to my profile
So that I can change my personal details.
4. **As a User**
I want to pay with multiple payment options
So that I can choose my preferred method.

Persona: Driver

User Stories:

5. **As a Driver**
I want to get a ride amount
So that I can check whether the ride is profitable.
6. **As a Driver**
I want to receive notifications after a user requests a ride
So that I can accept that ride.

7. As a Driver

I want to view ride details

So that I know the total distance of the ride

8. As a Driver

I want to update my ride status (e.g., started, completed)

So that users are informed of my progress.

Persona: Admin

User Stories:

9. As an Admin

I want to see my monthly earnings

So that I can balance my incomes.

10. As an Admin

I want to check user reviews

So that I can take actions on drivers based on feedback.

11. As an Admin

I want to check ongoing rides

So that I can check available drivers

12. As an Admin

I want to see how many drivers are registered

So that I can prepare reports.

Define the Scrum Roles

Product Owner

- Name: Sandamuthu A.A.N

Responsibilities:

- Define the product vision.
- Prioritize the product backlog.
- Ensure the team delivers value to the business.

Scrum Master

- Name: Janith Priyankara W.A

Responsibilities:

- Facilitate Scrum ceremonies (sprint planning, daily stand-ups, sprint reviews, and retrospectives).
- Remove impediments that may hinder the team's progress.
- Coach the team in Agile practices.

Development Team

- Developer 1: Madusanka W. A. L. P

Responsibilities: Focus on backend development (APIs, database management).

- Developer 2: Kulatunga W.M.H.P

Responsibilities: Work on frontend development (UI/UX design, client-side logic).

- Developer 3: Gamage G.A.G

Responsibilities: Handle testing and quality assurance (unit tests, integration tests).

- Developer 4: Sansiri P.S.T

Responsibilities: Assist deployment processes.

Prepare the Product Backlog

ID	User Story	Priority
PBI-01	As a User, I want to book a ride So that I can travel to my destination.	High
PBI-02	As a User, I want to see my rides so that I can view my ride history.	Low
PBI-03	As a User, I want to update my profile so that I can change personal details.	Low
PBI-04	As a User, I want to pay with multiple options.	Medium
PBI-05	As a Driver, I want to get a ride amount So that I can check whether ride is profitable or not	High
PBI-06	As a Driver, I want to get a notification after user request for a ride So that I can accept that ride	High
PBI-07	As a Driver, I want to get a ride detail So that I can view total distance of the ride	Medium
PBI-08	As a Driver, I want to update my ride status (e.g., started, completed) So that the system and user are aware of the current progress of the ride	Medium

PBI-09	As an Admin, I want to see my monthly earnings So that I can balance my incomes	High
PBI-10	As an Admin, I want to check user reviews and get actions for it So that I can remove and add drivers based on those reviews	Medium
PBI-11	As an Admin, I want to check ongoing rides So that I can check available drivers	High
PBI-12	As an Admin, I want to see how many drivers are registered So that I can get a report for HR	Low

Sprint Planning

1st Sprint – Week 4 to Week 7

Selected User Stories (according to their priorities): - ***Highlighted in yellow***

Sprint ID	User Story	Priority
SP1-01	As a User, I want to book a ride so that I can travel to my destination	High
SP1-02	As a Driver, I want to get a ride amount to check profitability.	High
SP1-03	As a Driver, I want to get a notification for ride requests.	High
SP1-04	As an Admin, I want to see my monthly earnings.	High
SP1-05	As an Admin, I want to check ongoing rides So that I can check available drivers	High

Sprint Backlog

Task Breakdown for User Story SP1-01 and Time Estimation (Ride Booking Functionality)

Task ID	Task	Est.hours
T-01.1	Design UI for ride booking page	6
T-01.2	Implement backend API for ride booking	8
T-01.3	Validate user input for booking	4
T-01.4	Integrate payment gateway for booking	7
T-01.5	Create confirmation system for booked rides	5
T-01.6	Write unit tests for booking functionality	4
T-01.7	Conduct user testing for ride booking	5

Task Breakdown for User Story SP1-02 and Time Estimation (Ride Amount Calculation)

Task ID	Task	Est.hours
T-02.1	Design UI for displaying ride amounts	4
T-02.2	Implement backend logic for ride amount calculation	6
T-02.3	Create functionality to show estimated earnings	5
T-02.4	Test ride amount calculation functionality	4

Task Breakdown for User Story SP1-03 and Time Estimation (Ride Request Notifications)

Task ID	Task	Est.hours
T-03.1	Design notification UI for ride requests	4
T-03.2	Implement notification system for ride requests	6
T-03.3	Set up notifications	7
T-03.4	Test notification functionality	4

Task Breakdown for User Story SP1-04 and Time Estimation (Monthly Earnings Overview)

Task ID	Task	Est.hours
T-04.1	Design admin dashboard for monthly earnings	5
T-04.2	Implement backend logic for calculating monthly earnings	6
T-04.3	Create functionality to display earnings in charts	7
T-04.4	Test monthly earnings functionality	4

Task Breakdown for User Story SP1-05 and Time Estimation (Check Ongoing Rides)

Task ID	Task	Est.hours
T-05.1	Design admin interface for viewing ongoing rides	5
T-05.2	Implement backend API for retrieving ongoing rides	6
T-05.3	Create functionality for displaying available drivers	5
T-05.4	Test ongoing rides functionality	4

2nd Sprint – Week 7 to week 9

Sprint ID	User Story	Priority
SP2-01	As a User, I want to pay with multiple options.	Medium
SP2-02	As a Driver, I want to get a ride detail So that I can view total distance of the ride	Medium
SP2-03	As a Driver, I want to update my ride status (e.g., started, completed) So that the system and user are aware of the current progress of the ride	Medium
SP2-04	As an Admin, I want to check user reviews and get actions for it So that I can remove and add drivers based on those reviews	Medium

Sprint Backlog

Task Breakdown for User Story SP2-01 and Time Estimation (Multiple Payment Options)

Task ID	Task	Est.hours
T-01.1	Design UI for payment options	5
T-01.2	Implement backend logic for multiple payment methods	8
T-01.3	Integrate payment gateways (e.g., credit card, PayPal)	7
T-01.4	Create functionality for handling payment confirmations	4
T-01.5	Test payment processing functionality	5

Task Breakdown for User Story SP2-02 and Time Estimation (Ride Detail for Drivers)

Task ID	Task	Est.hours
T-02.1	Design UI for displaying ride details	4
T-02.2	Implement backend API for retrieving ride details	6
T-02.3	Calculate total distance for the ride	5
T-02.4	Test ride details functionality	4

Task Breakdown for User Story SP2-03 and Time Estimation (Update Ride Status)

Task ID	Task	Est.hours
T-02.1	Design UI for updating ride status	4
T-02.2	Implement backend API for updating ride status	6
T-02.3	Create notifications for status updates	5
T-02.4	Test ride status update functionality	4

Task Breakdown for User Story SP2-04 and Time Estimation (Check User Reviews)

Task ID	Task	Est.hours
T-03.1	Design admin interface for viewing user reviews	5
T-03.2	Implement backend logic for retrieving reviews	6
T-03.3	Create functionality for taking actions on reviews	7
T-03.4	Test review management functionality	4

3rd Sprint – Week 9 to week 11

Sprint ID	User Story	Priority
SP3-01	As a User, I want to see my rides so that I can view my ride history.	Low
SP3-02	As a User, I want to update my profile so that I can change personal details.	Low
SP3-03	As a Driver, I want to update my ride status (e.g., started, completed) So that the system and user are aware of the current progress of the ride	Low
SP3-04	As an Admin, I want to see how many drivers are registered So that I can get a report for HR	Low

Sprint Backlog

Task Breakdown for User Story SP3-01 (View Ride History)

Task ID	Task	Est.hours
T-01.1	Design UI for viewing ride history	4
T-01.2	Implement backend API for retrieving ride history	6
T-01.3	Create functionality to filter and sort rides	5
T-01.4	Test ride history retrieval functionality	4

Task Breakdown for User Story SP3-02 (Update User Profile)

Task ID	Task	Est.hours
T-02.1	Design UI for profile update page	4
T-02.2	Implement backend API for updating user profile	6
T-02.3	Validate user inputs for profile updates	3
T-02.4	Test profile update functionality	4

Task Breakdown for User Story SP3-03 (Update Ride Status)

Task ID	Task	Est.hours
T-03.1	Design UI for updating ride status	4
T-03.2	Implement backend API for updating ride status	6
T-03.3	Create notifications for status updates	5
T-03.4	Test ride status update functionality	4

Task Breakdown for User Story SP3-04 (View Registered Drivers)

Task ID	Task	Est.hours
T-04.1	Design UI for updating ride status	4
T-04.2	Implement backend API for updating ride status	6
T-04.3	Create notifications for status updates	5
T-04.4	Test ride status update functionality	4